

CMVS – lab1

Simulation of LTI systems (1)

- Consider a LTI system represented by

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) \end{cases}$$

- To simulate its behavior with a known input $\mathbf{u}(t)$, we can use the closed form solution of the system,

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)}\mathbf{x}(t_0) + \int_{t_0}^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau$$

Simulation of LTI systems (2)

$$\mathbf{x}(t) = e^{\mathbf{A}(t-t_0)}\mathbf{x}(t_0) + \int_{t_0}^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau$$

- The first term can be directly computed.
- The integral in the second term can be evaluated symbolically or numerically.
- Let us do it numerically: approximate it as follows using Euler integration (assuming a constant integration step Δt),

$$\int_{t_0}^{t_k} e^{\mathbf{A}(t_k-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau \approx \sum_{i=0}^{k-1} e^{\mathbf{A}(t_k-t_i)}\mathbf{B}\mathbf{u}(t_i) \Delta t$$

Simulation of LTI systems (3)

- A different way of simulating the system consists in directly integrating the state equations.
- Using Euler integration we get

approximation of the derivative

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t (\mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k)$$

Exercise: ROV with 3 dofs

Equations of motion:

one equation for every dof

$$m\ddot{x} = -\alpha\dot{x} + F_x$$

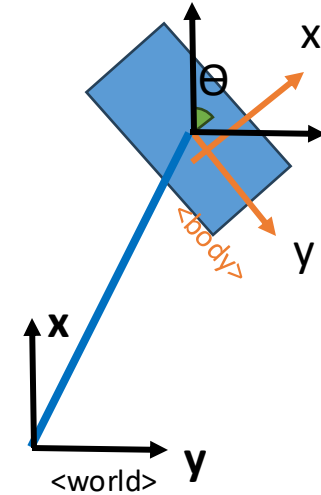
$$m\ddot{y} = -\beta\dot{y} + F_y$$

$$J\dot{w} = -\gamma w + N_z$$

$f = ma$

where:

- m = vehicle mass
- J = vehicle inertia on z axis
- α, β, γ are the drag coefficients
- F_x, F_y are the forces on x, y axes
- N_z is the torque around z axis



Questions:

Q1. Choose a convenient state vector and write the equation of motion in the general structure of LTI representation.

Q2. Define the system in the MATLAB script *lab2.m*: state vector, parameters and matrices.

Q3. Estimate the state evolution $\mathbf{x}(t)$, using the analytical formula and assuming a constant input vector. Input vector and initial state are already defined in *lab2.m*. To do this, fill the function *SimulateSysAnalytical* given in the template.

Q4. Repeat Q3 using Euler integration by filling the function *SimulateSysEuler*.

Q5. Plot the state estimated by the two methods and the percentage of difference between the estimates (follow the template given in *lab2.m*).

Q6. Briefly discuss the different between the two estimates.

Q7. Briefly discuss possible outputs.